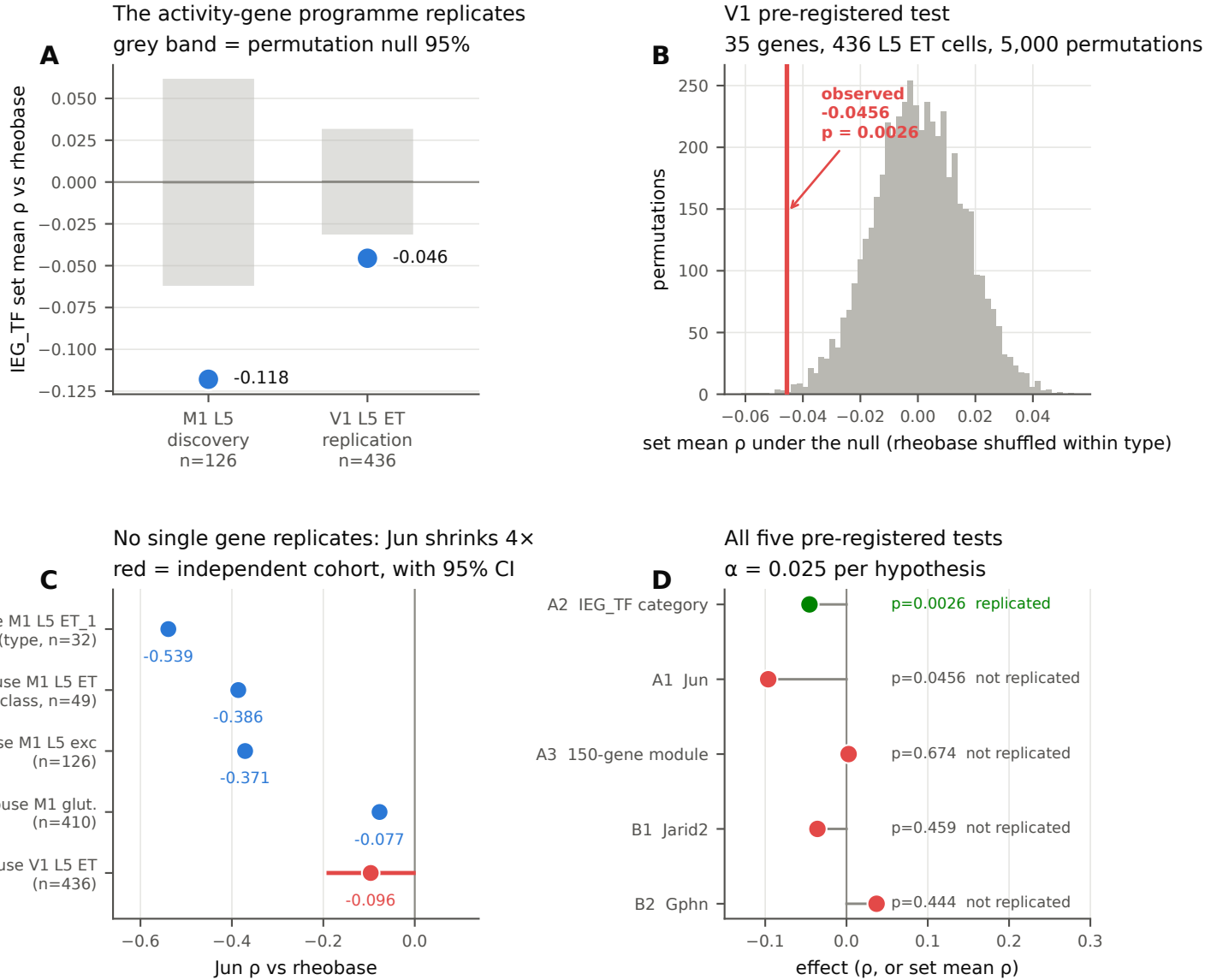
A
The activity-gene programme replicates
grey band = permutation null 95%
IEG_TF set mean ρ vs rheobase
0.050
0.025
0.000
−0.025
−0.050
−0.075
−0.100
−0.125
-0.046
-0.118
M1 L5
discovery
n=126
V1 L5 ET
replication
n=436
B
V1 pre-registered test
35 genes, 436 L5 ET cells, 5,000 permutations
permutations
250
200
150
100
50
0
observed
-0.0456
p = 0.0026
−0.06 −0.04 −0.02 0.00 0.02 0.04
set mean ρ under the null (rheobase shuffled within type)
C
No single gene replicates: Jun shrinks 4×
red = independent cohort, with 95% CI
mouse M1 L5 ET_1
(type, n=32)
mouse M1 L5 ET
(class, n=49)
mouse M1 L5 exc
(n=126)
mouse M1 glut.
(n=410)
mouse V1 L5 ET
(n=436)
-0.539
-0.386
-0.371
-0.077
-0.096
−0.6 −0.4 −0.2 0.0
Jun ρ vs rheobase
D
All five pre-registered tests
α = 0.025 per hypothesis
A2 IEG_TF category
A1 Jun
A3 150-gene module
B1 Jarid2
B2 Gphn
p=0.0026 replicated
p=0.0456 not replicated
p=0.674 not replicated
p=0.459 not replicated
p=0.444 not replicated
−0.1 0.0 0.1 0.2 0.3
effect (ρ, or set mean ρ)